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ICON Central Office - Madhapur - Hyderabad

Sec: Sr.Super60_NUCLEUS & STERLING_BT

Paper -1 (Adv-2022-P1-Model)

Date: 21-07-2024

Time: 09.00Am to 12.00Pm

RPTA-02

Max. Marks: 180

21-07-2024_Sr.Super60_NUCLEUS&STERLING_BT_Jee-Adv(2022-P1)_RPTA-02_Syllabus

PHYSICS

: Thermal Physics: Second law of thermodynamics, Reversible and irreversible processes, Carnot engine and its efficiency (Important for ADVANCED);

Heat Transfer: Heat conduction in one dimension; Elementary concepts of convection and radiation; Blackbody radiation: absorptive and emissive powers; Kirchhoff's law; Wien's displacement law, Stefan's law, Newton's law of cooling.

Ray Optics: Rectilinear propagation of light; Reflection, Magnification.

CHEMISTRY

: GOC: Inductive effect, Resonance and hyperconjugation; Keto-enol tautomerism; Hydrogen bonding- definition and their effects on physical properties of alcohols and carboxylic acids; Inductive and resonance effects on acidity and basicity of organic acids and bases; Polarity and inductive effects in alkyl halides; Reactive intermediates produced during homolytic and heterolytic bond cleavage; Formation, structure and stability of carbocations, carbanions and free radicals

Alkanes: Preparation, properties and reactions of alkanes. Homologous series, physical properties of alkanes (melting points, boiling points and density) and effect of branching on them; Combustion and halogenations of alkanes (including allylic and benzylic halogenation); Preparation of alkanes by Wurtz reaction and decarboxylation reaction, Corey-House Reaction.

MATHEMATICS : LCD

Name of the Student: _____

H.T. NO:

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**JEE-ADVANCE-2022-P1-Model**

Time:3Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 180

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Matching Type	+3	-1	4	12
Total				18	60

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Matching Type	+3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Matching Type	+3	-1	4	12
Total				18	60

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**MATHEMATICS****Max Marks: 60****SECTION – I
(NUMERICAL VALUE TYPE)**This section contains **EIGHT (08)** questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the correct numerical value is entered ;

Partial Mark: 0 In all other cases.

1. Let $f : (0, \infty) \rightarrow (0, \infty)$ be a function differentiable at $x = 3$ and satisfying

$$f(3) = 3 \cdot f'(3). \text{ If the value of } \lim_{x \rightarrow \infty} \left(\frac{f\left(3 + \frac{3}{x}\right)}{f(3)} \right)^x \text{ is } L \text{ then } [L] \text{ is equal to } ([] \text{ denotes}$$

greatest integer function)

2. Let $f(x) = \begin{cases} \{x^3\}, & -1 \leq x < 1 \\ |x-2|, & 1 \leq x < 3 \\ \operatorname{sgn}(x^2 - x - 1), & 3 \leq x \leq 4 \end{cases}$ (where $\{.\}$ denotes the fractional Part

function) if n_1 is the number of points of discontinuity of $f(x)$ in $[-1, 4]$ and n_2 is the number of points of non differentiability in $(-1, 4)$ then $n_1 \cdot n_2$ is equal to

3. If $f(x) = 3(1 + |\tan x|)^{\frac{\alpha}{|\tan x|}}, \frac{-1}{2} < x < 0$
 $= \beta$ $x = 0$

$$= 3 \left(1 + \left| \frac{\sin x}{3} \right| \right)^{\frac{6}{|\sin x|}}, 0 < x < \frac{2}{3}$$

Is a continuous function at $x = 0$, then the value of $\alpha + \frac{\beta}{e^2}$ is equal to

4. If $L = \lim_{x \rightarrow 0} \frac{\sin x^4 - x^4 \cos x^4 + 4x^{20}}{x^4(e^{2x^4} - 1 - 2x^4)}$, then the value of $\frac{1}{L}$ is

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5. If $L = \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{2^r}{5^{2^r} + 1}$ then the value of $4L$ is equal to
6. A continuous function $f(x) = \begin{cases} x+a, & \text{for } |x| < 2 \\ b f\left(\frac{x}{2}\right) + c, & \text{for } |x| \geq 2 \end{cases}$ is defined for some non-zero constants a, b and c . Then the value of $\frac{a}{c} + b$ is
7. If $\lim_{\alpha \rightarrow 0} \frac{e^{\cos(\alpha^n)} - e}{\alpha^m} = \frac{-e}{2}$ where m and n positive integers greater than 1, Then the value of $\frac{m}{n}$ is _____
8. Consider a function $f: R \rightarrow R$ such that $f(x) = \begin{cases} \sin(\pi x), & \text{if } x \in Q \\ \tan(\pi \sqrt{|x|}), & \text{if } x \notin Q \end{cases}$. If $\lim_{x \rightarrow N} f(x)$ exists, then the sum of all positive integers $N < 100$, is equal to

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;

Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

9. The function $f(x) = \begin{cases} 5x-4, & \text{for } 0 < x \leq 1 \\ 4x^2-3x, & \text{for } 1 < x < 2 \\ 3x+4, & \text{for } x \geq 2 \end{cases}$ is

- A) continuous at $x=1$ and $x=2$
 B) continuous at $x=1$ but not derivable at $x=2$
 C) continuous at $x=2$ but not derivable at $x=1$
 D) continuous at $x=1$ and $x=2$ but not derivable at $x=1$ and $x=2$

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10. If $\lim_{x \rightarrow 1} (2 - x + a[x - 1] + b[1 + x])$ exists, then a and b can take the values (where $[.]$ denotes the greatest integer function)
- A) $a = 1/3, b = 1$ B) $a = 1, b = -1$ C) $a = 9, b = -9$ D) $a = 2, b = 2/3$
11. If $A = \lim_{x \rightarrow 0} \frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ and $B = \lim_{x \rightarrow 0} \frac{[x]}{x}$, then which of the following is/are true?
- ($[.]$ is G.I.F)
- A) $A = 1$ B) A does not exist
C) $B = 0$ D) $B = 1$
12. Let $g(x) = \frac{(x-1)^n}{\log \cos^m(x-1)}$, $0 < x < 2$, where m and n are integers such that $m \neq 0, n > 0$. If $\lim_{x \rightarrow 1^+} g(x) = -1$, then which of the following is/are true?
- A) $m = 1$ B) $n = -1$ C) $m = 2$ D) $n = 2$
13. If $h(x) = \lim_{n \rightarrow \infty} \frac{x^n f(x) + g(x) + 3}{2x^n + 4x + 1}$ when $x \neq 1$ and $h(1) = e^2$. Such that $f(x), g(x), h(x)$ are continuous function at $x = 1$ then which of the following is/are true?
- A) $f(1) = 2e^2$ B) $g(1) = 5e^2 - 3$
C) $f(1) + g(1) = 7e^2 + 5$ D) $f(1) - g(1) = 7e^2 + 5$
14. If $y = f(x)$ is a function satisfying the relation $y^2 - x^2 y = x$, then which of the following is/are true?
- A) $f'(x) = \frac{1 + 2x f(x)}{2f(x) - x^2}$ B) $f'(x) = \frac{f(x) + 2x f^2(x)}{f^2(x) + x}$
C) $f'(1) = 1 + \frac{2}{\sqrt{5}}$ D) $f'(1) = 1 - \frac{2}{\sqrt{5}}$





SECTION – III (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks: +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -1 In all other cases.

15. Match the following

Column I

Column II

A) $\lim_{x \rightarrow 2^+} \{x\} \frac{\sin(x-2)}{(x-2)^2} =$

p) -3

B) $\lim_{x \rightarrow 2} [2-x] + [x-2] - x =$

q) 1

C) $\lim_{x \rightarrow 2} \{2-x\} + \{x-2\} =$

r) -1

D) $\lim_{x \rightarrow 0^+} \frac{\frac{1}{e^x} - \frac{-1}{e^x}}{\frac{1}{e^x} + \frac{-1}{e^x}} =$

s) 2

(Note: [] is G.I.F and { } is fractional part of function)

A) $A \rightarrow q, B \rightarrow p, C \rightarrow q, D \rightarrow q$

B) $A \rightarrow pr, B \rightarrow r, C \rightarrow p, D \rightarrow q$

C) $A \rightarrow p, B \rightarrow q, C \rightarrow rs, D \rightarrow p$

D) $A \rightarrow q, B \rightarrow pq, C \rightarrow r, D \rightarrow s$

16. Match the items given under List I with the items given under List II

List I

List II

1) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\left(\frac{\pi}{2} - x\right) \sin x - 2 \cos x}{\left(\frac{\pi}{2} - x\right) + \cot x} =$

i) -1

2) $2^{1/4} \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{\cos x} - \sqrt{\sin x}}{x - \pi/4} =$

ii) 1/4

3) $8 \lim_{x \rightarrow \frac{\pi}{8}} \frac{\cot 4x - \cos 4x}{(\pi - 8x)^3} =$

iii) -1/2

4) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sqrt{2 - \sin x} - 1}{\left(\frac{\pi}{2} - x\right)^2} =$

iv) 1/2

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- A) (1)-(iii), (2)-(iv), (3)-(i), (4)-(ii) B) (1)-(ii), (2)-(iii), (3)-(iv), (4)-(i)
 C) (1)-(iii), (2)-(i), (3)-(iv), (4)-(ii) D) (1)-(iv), (2)-(i), (3)-(ii), (4)-(iii)

17. Let $f(x) = 3x^3 + bx^2 + bx + 3, b \in R$. If $x = 2$ is a root of the equation $f(x) = 0$, then match the List I with the List II.

List I

List II

1) $\lim_{x \rightarrow 2} \frac{e^{f(x)} - 1}{x - 2} =$

i) 27

2) $\lim_{x \rightarrow -1} \frac{\sin(2f(x))}{x + 1} =$

ii) -27/8

3) $\lim_{x \rightarrow 1/2} \frac{\log_e(1 + f(x))}{2x - 1} =$

iii) 54

4) If $\lim_{x \rightarrow -1} (1 + 4f(x))^{\frac{1}{x+1}} = e^k$, then $k =$

iv) 27/2

- A) (1)-(iii), (2)-(iv), (3)-(i), (4)-(ii) B) (1)-(ii), (2)-(iii), (3)-(iv), (4)-(i)
 C) (1)-(iii), (2)-(i), (3)-(iv), (4)-(i) D) (1)-(iv), (2)-(i), (3)-(ii), (4)-(iii)

18. For different functions given under the List I match the corresponding number of points of non-differentiability given under the list II

List I

List II

1) $f(x) = |x^2 - 1| \cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$

i) 0

2) $f(x) = (\cos x) \cdot \sin^{-1}(\sin x)$

ii) 1

3) $f(x) = |2x^2 - x - 3| (\sin \pi x)$

iii) 2

4) $f(x) = x^{2/3} (x - 1)^{4/7}$

iv) 3

- A) (1)-(iii), (2)-(iv), (3)-(i), (4)-(ii) B) (1)-(ii), (2)-(iii), (3)-(iv), (4)-(i)
 C) (1)-(iii), (2)-(i), (3)-(iv), (4)-(ii) D) (1)-(iv), (2)-(i), (3)-(ii), (4)-(iii)



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PHYSICS

Max Marks: 60

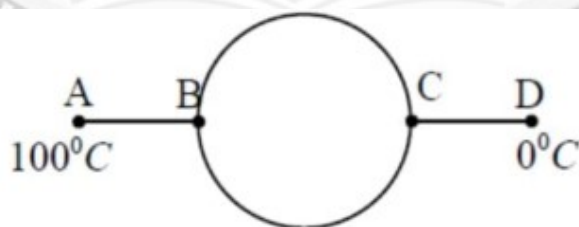
SECTION – I
(NUMERICAL VALUE TYPE)This section contains **EIGHT (08)** questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the correct numerical value is entered ;

Partial Mark: 0 In all other cases.

19. Two spherical stars A and B emit blackbody radiation. The radius of A is 400 times that of B and A emits 10^4 times the power emitted from B. The ratio (λ_A / λ_B) for their wavelengths λ_A and λ_B at which the peaks occur in their respective radiation curves is :
20. A metal rod AB of length $10x$ has its one end A in ice at 0°C , and the other end B in water at 100°C . If a point P on the rod is maintained at 40°C , then it is found that equal amounts of water and ice evaporate and melt per unit time. The latent heat of evaporation of water is 540 cal/g and latent heat of melting of ice is 80 cal/g . If the point P is at a distance of λx from the ice end A, find the value λ . [Neglect any heat loss to the surrounding.]
21. Two identical conducting rods AB and CD are connected to a circular conductor's diametrically opposite points B and C. The radius of the ring is equal to the length of rods. The area of cross-section, and thermal conductivity of the rod and ring are equal. The ends A & D are maintained at temperatures of 100°C and 0°C . Temperature of point C will be



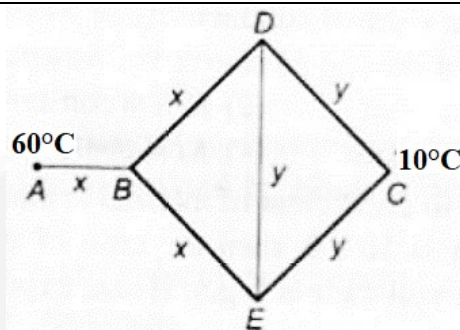
22. Consider the 6 identical rods as shown in figure, The conductivity of x is double that of y . The temperature of the junction B is ____ ($^\circ\text{C}$)

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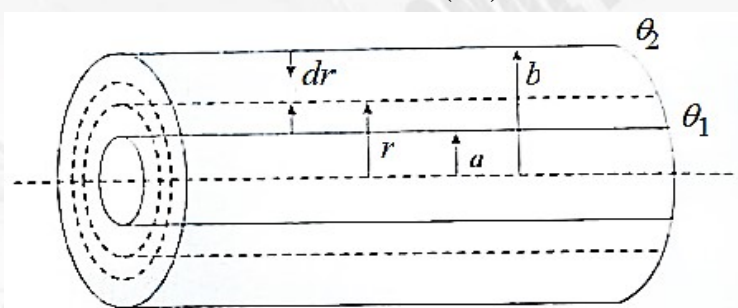
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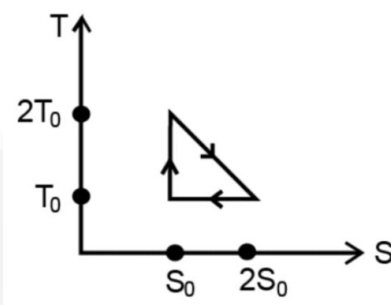
23. A 2 m long wire of resistance 4 ohm and diameter 0.64 mm is coated with plastic insulation of thickness 0.06mm. When a current of 5 ampere flows through the wire, find the temperature difference across the insulation in steady state ($^{\circ}\text{C}$) if

$[K = 0.16 \times 10^{-2} \text{ cal / cm} - ^{\circ}\text{C} - \text{sec}]$. (Take $\ln(38) = 3.64$)



24. One end of a rod of length 20cm is inserted in a furnace at 800K. The sides of the rod are covered with an insulating material and the other end emits radiation like a black body. The temperature of this end is 750 K in the steady state. The temperature of the surrounding air is 300K. Assuming radiation to be the only important mode of energy transfer between the surroundings and the open end of the rod, find the thermal conductivity of the rod. Stefan constant $\sigma = 6.0 \times 10^{-8} \text{ W / m}^2 \cdot \text{K}^4$. Take emissivity of the open end $e = 1$ (in w/m-K)
25. A solid black sphere of density ' ρ ', specific heat capacity ' C ' and radius ' R ' is at an initial temperature 200K. On allowing it to cool under the process of radiation then the time taken for it to cool to 100K is found to be $\frac{7R\rho C}{8K\sigma}$, micro second . If ' σ ' is Stefan's constant, then find the value of ' K '. Assume the surrounding temperature to be extremely small and emissivity of the outer surface of the sphere to be unity.
26. The temperature – entropy diagram of a reversible engine cycle is given in the figure. Its efficiency is $k/3$. value of k





SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

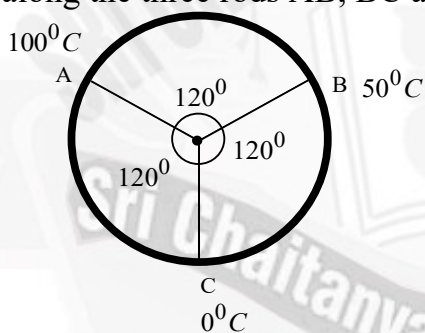
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

Partial Marks : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

27. A Circular ring (centre O) of radius a , and of uniform cross section is made up of three different metallic rods AB, BC and CA (joined together at the points A, B and C in pairs) of thermal conductivities α_1, α_2 and α_3 respectively (see diagram). The junction A, B and C are maintained at the temperatures 100°C , 50°C and 0°C , respectively, All the rods are of equal lengths and cross sections. Under steady state conditions, assume that no heat is lost from the sides of the rods. Let Q_1, Q_2 and Q_3 be the rates of transmission of heat along the three rods AB, BC and CA. Then



- A) $Q_1 = Q_2 = Q_3$ and all are transmitted in the clockwise sense
 B) Q_1 and Q_2 flow in clockwise sense and Q_3 in the anticlockwise sense.
 C) $Q_1 : Q_2 : Q_3 :: \alpha_1 : \alpha_2 : 2\alpha_3$
 D) $\frac{Q_1}{\alpha_1} + \frac{Q_2}{\alpha_2} = \frac{Q_3}{\alpha_3}$


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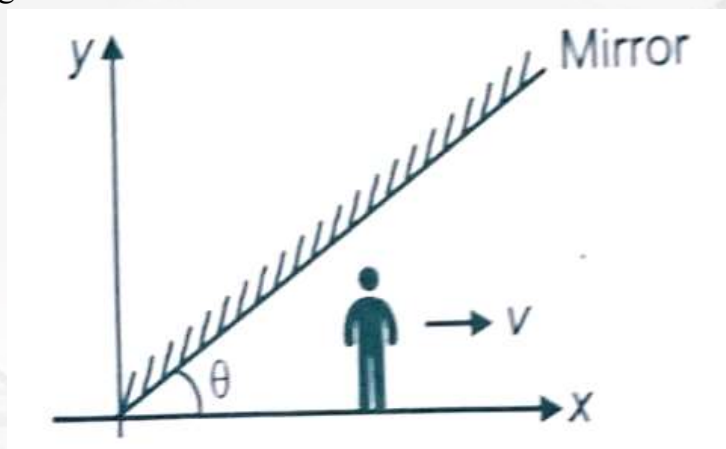
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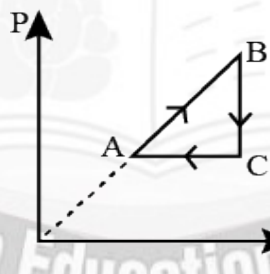
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28. Four Carnot engines operate between reservoir temperature of (i) 300K and 400K (ii) 400K and 500K (iii) 500 K and 600 K (iv) 600 K and 800 K which engines has greatest thermal efficiency?
- A) i B) ii C) iii D) iv
29. A boy is walking under an inclined mirror at a constant velocity v m/s along x-axis as shown in figure. If the mirror is inclined at an angle θ with the horizontal then the velocity of image is



- A) $v \sin \theta \hat{i} + v \cos \theta \hat{j}$ B) $v \cos \theta \hat{i} + v \sin \theta \hat{j}$
 C) $v \sin 2\theta \hat{i} + v \cos 2\theta \hat{j}$ D) $v \cos 2\theta \hat{i} + v \sin 2\theta \hat{j}$
30. Consider a thermodynamic cycle in a PV diagram shown in the figure Performed on one mole of a ideal monoatomic gas. The temperature at A is T_0 and volume at A and B are related as $V_B = V_C = 2V_A$. Choose the correct option(s) from the following.

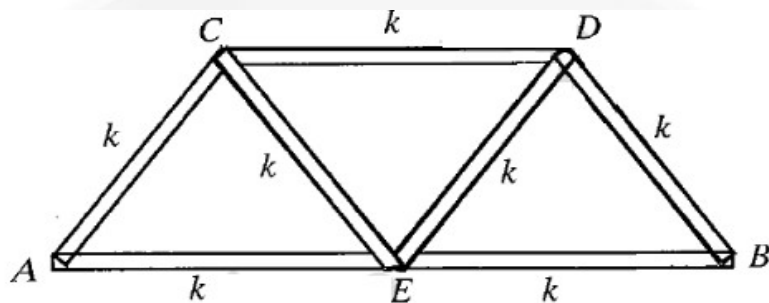


- A) The maximum temperature during the cycle is $4T_0$
 B) Net work done by the gas during the cycle is $0.5 RT_0$
 C) The heat capacity of the process AB is $2R$
 D) The efficiency of the cycle is 8.33%

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31. Seven identical rods of material of thermal conductivity k are connected as shown in fig. All the rods are of identical length l and cross sectional area A if the one end B is kept at 100°C and the other end is kept at 0°C what would be the temperatures of the junctions C, D and E (θ_C, θ_D) and (θ_E) in the steady state?



- A) $\theta_C > \theta_E > \theta_D$
 B) $\theta_E = 50^\circ\text{C}$ and $\theta_D = 37.5^\circ\text{C}$
 C) $\theta_E = 50^\circ\text{C}$ and $\theta_C = 62.5^\circ\text{C}$ and $\theta_D = 37.5^\circ\text{C}$
 D) $\theta_E = 50^\circ\text{C}$ and $\theta_C = 60^\circ\text{C}$ and $\theta_D = 40^\circ\text{C}$
32. Metal cylinder of mass 0.5kg is heated electrically by a 12W heater in a room at 15°C . The cylinder temperature rises uniformly to 20°C in 5min and finally becomes constant at 45°C . Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings:
- A) The rate of loss of heat of the cylinder to surrounding at 20°C is 2W
 B) The rate of loss of heat of the cylinder to surrounding at 45°C is 12W
 C) The rate of loss of heat of the cylinder to surrounding at 20°C is 5W
 D) The rate of loss of heat of the cylinder to surrounding at 45°C is 30W

SECTION – III (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -1 In all other cases.



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33. A rod AB of uniform cross-section consists of four section AC, CD, DE and EB of different metals with thermal conductivities K , $(0.8)K$, $(1.2)K$ and $(1.50)K$, respectively. Their lengths are respectively L , $(1.2)L$, $(1.5)L$ and $(0.6)L$. They are joined rigidly in succession at C,D and E to form the rod AB, The end A is maintained at $100^\circ C$ and the end B is maintained at $0^\circ C$. The steady state temperatures of the joints C,D and E are respectively T_C, T_D and T_E . Column I lists the temperature difference $(T_A - T_C), (T_C - T_D), (T_D - T_E)$ and $(T_E - T_B)$ in the four sections and column II their values jumbled up. Match each item in column I with its correction value in column II.



Column I		Column II	
I.	$(T_A - T_C)$	P.	9.6
II.	$(T_C - T_D)$	Q.	30.1
III.	$(T_D - T_E)$	R.	24.1
IV.	$(T_E - T_B)$	S.	36.2

A) I-R ; II -S ; III-Q ; IV-P

B) I-P ; II -Q ; III-S; IV-R

C) I-R ; II -S ; III-P ; IV-Q

D) I-R ; II -Q ; III-P ; IV-S

34. Match the temperature of a black body given in List-I with an appropriate statement in List-II, and choose the correct option

[Given : Wien's constant as $2.9 \times 10^{-3} m - K$ and $\frac{hc}{e} = 1.24 \times 10^{-6} V - m$]

List- I		List- II	
I.	2000K	P.	The radiation at peak wavelength can lead to emission of photoelectrons from a metal of work function 4eV.
II.	3000K	Q.	The radiation at peak wavelength is visible to human eye.
III.	5000K	R.	The radiation at peak emission wavelength will result in the widest central maximum of a singles slit diffraction.
IV.	10000K	S.	The power emitted per unit area is 1/16 of the emitted by a blackbody at temperature 6000K.
		T.	The radiation at peak emission wavelength can be used to image human bones.



- D) I-P; II-Q ;III-T ; IV-R**

	List- I		List- II
I.	An object is placed at a distance equal to focal length from pole in front of convex mirror	P.	Magnification is- ∞
II.	An object is placed at centre of curvature in front of a concave mirror(B)	Q.	Magnification is 0.5
III.	An Object is Placed at Focus in front of a concave mirror(C)	R.	Magnification is +1
IV.	An object is placed at a distance equal to radius of curvature in front of a convex mirror (D)	S.	Magnification is -1
		T	Magnification is + 1/3

- D) I-R; II-T ;III-Q ; IV-S**

List- I		List- II	
I.	Temperature b	P.	$10^{\circ}C$
II.	Temperature c	Q.	$40^{\circ}C$
III.	Temperature f	R.	$-20^{\circ}C$
IV.	Temperature d	S.	None of these

- D) I-S; II-P; III-R ; IV-P**



CHEMISTRY

Max Marks: 60

SECTION – I
(NUMERICAL VALUE TYPE)

This section contains EIGHT (08) questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 ONLY if the correct numerical value is entered ;

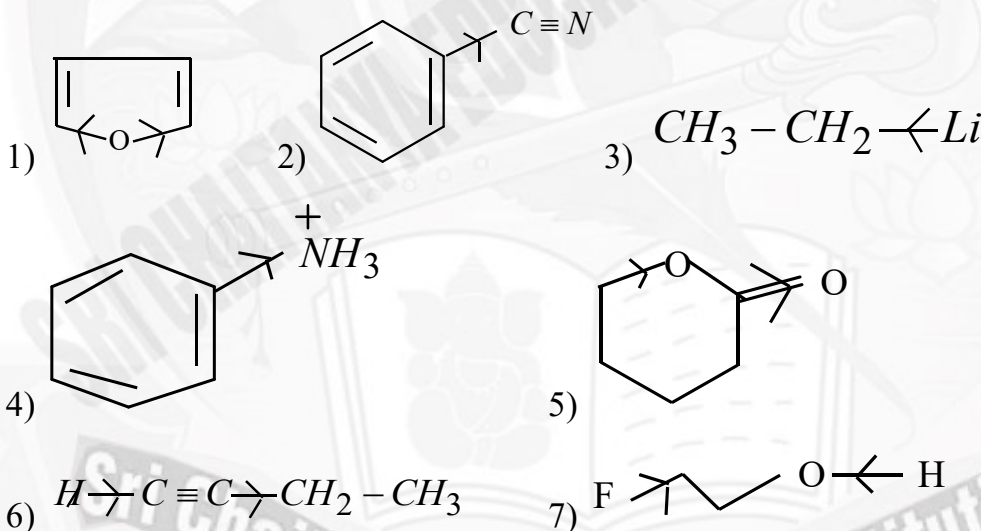
Partial Mark: 0 In all other cases.

37. The maximum number of possible isomers (including stereoisomers) which may be formed on mono-Chlorination of 1-methylcyclohex-1-ene using Cl_2 and UV light is

38. How many of the following compounds have intra molecular H-bonding ?

- | | |
|---------------------|---------------------------|
| 1) o-nitrophenol | 2) o-hydroxy benzoate ion |
| 3) o-hydroxy phenol | 4) Maleic acid |
| 5) Fumaric acid | 6) Chloral hydrate |
| 7) Acetone | 8) Fluoro benzene |
| | 9) Ethylacetate |

39. In how many structures the direction of inductive effect is correctly matched ?



40. How many of the following are ambident nucleophiles ?



41. The positive charge on the carbon can be delocalized over how many O atoms in the resonating structures ?

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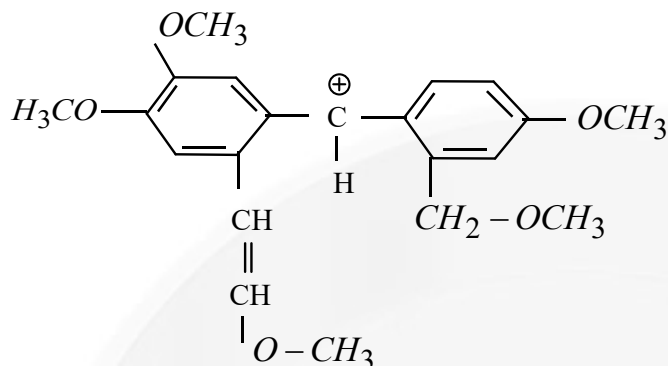
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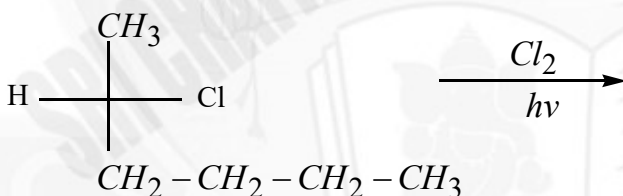
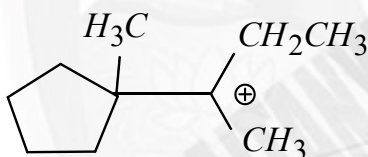
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42. Number of compounds which are more basic than aniline is
- 1) N-Methyl aniline
 - 2) N-methyl ethanamine
 - 3) Acetanilide
 - 4) o-Toluidine
 - 5) 4-methylaniline
 - 6) Benzyl amine
 - 7) Ammonia
 - 8) Benzamide
43. The total number of contributing structures showing the hyperconjugation involving C-H bonds for the following carbocation is



44. Total number of products obtained by mono chlorination of (S) -2- chlorohexane (including stereoisomers) is

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;

Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

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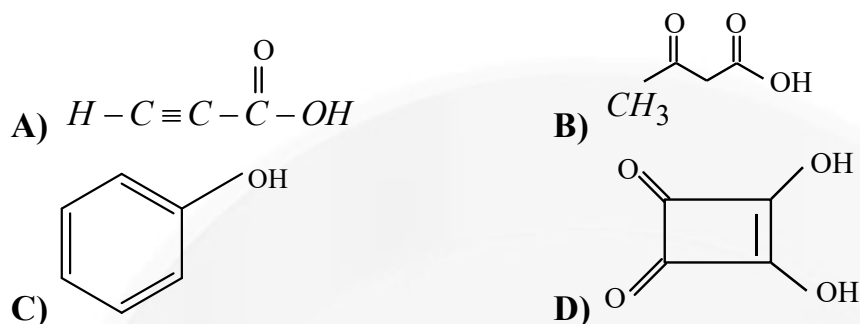


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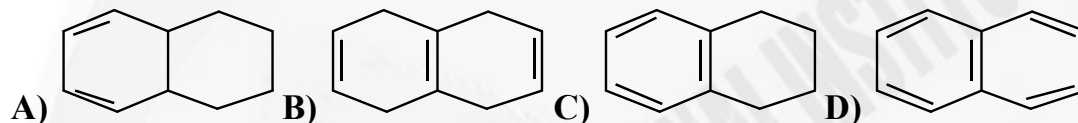
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45. Compounds which can liberate CO_2 on reacting with $NaHCO_3$ is/are

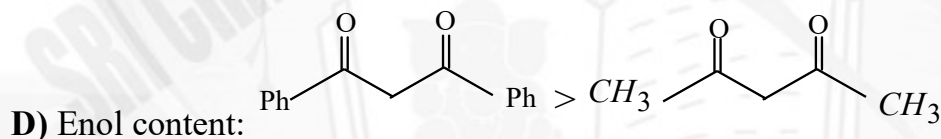
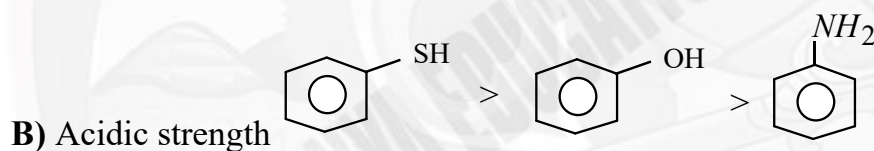


46. The compounds which are stabilized by resonance is/are

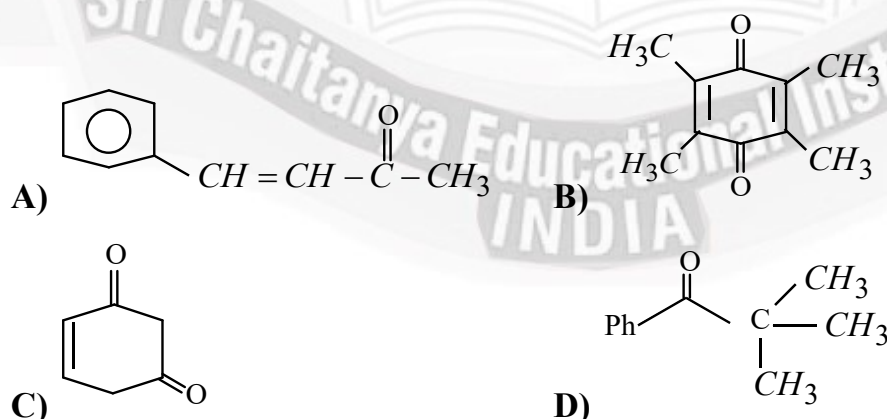


47. Which of the following is/are correct

A) Basic strength: $F^- > Cl^- > Br^- > I^-$



48. Tautomerism is exhibited by



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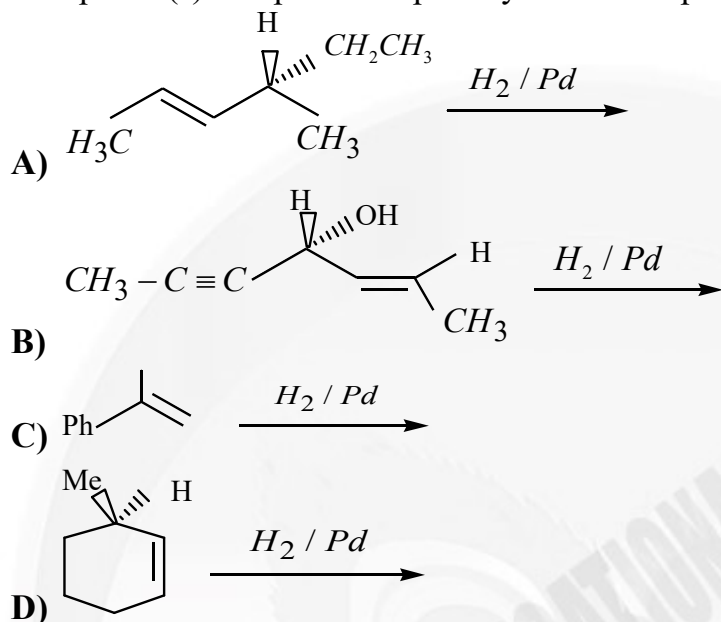
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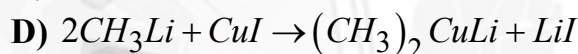
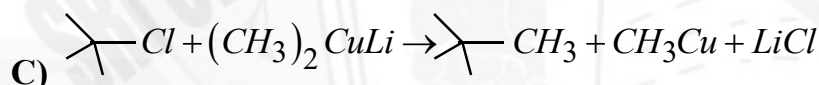
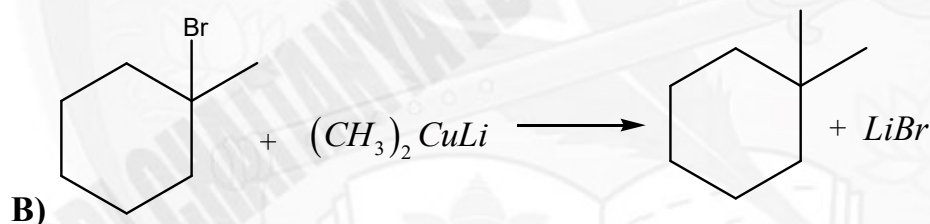
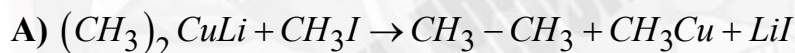
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49. Compound(s) that produce optically active compounds during reduction is/are



50. Which of the following reactions is/are correct



SECTION – III (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
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Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -1 In all other cases.





51. List I contains the reactants & List – II contains reagent

- | List – I | List –II |
|---|---|
| I) $\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{Br} \longrightarrow \text{CH}_3-\text{C}(\text{CH}_3)_2-\text{H}$ | P) H_2 / Ni |
| II) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}_2 \longrightarrow \text{CH}_3-\text{C}(\text{H})(\text{CH}_3)-\text{CH}_3$ | Q) NaBH_4 |
| III) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}_2 \longrightarrow \text{CH}_3-\text{C}(\text{CH}_2\text{Br})=\text{CH}_2$ | R) i) LiAlH_4 / ii) H_2O |
| IV) $\text{Cyclopropane} \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_3$ | S) $\text{NBS} / h\nu$ |
- A) $I \rightarrow R, II \rightarrow P, III \rightarrow Q, IV \rightarrow S$ B) $I \rightarrow Q, II \rightarrow P, III \rightarrow S, IV \rightarrow P$
 C) $I \rightarrow Q, II \rightarrow P, III \rightarrow S, IV \rightarrow Q$ D) $I \rightarrow R, II \rightarrow Q, III \rightarrow S, IV \rightarrow P$

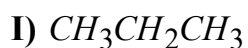
52. List –I contains reactants and List-II contains products

- | List-I | List-II |
|--|--|
| I) $\text{Br}-\text{C}_6\text{H}_{11}-\text{Br} \longrightarrow \text{Cyclohexane}$ | P) Red P+HI |
| II) $\text{Cyclopentanone} \longrightarrow \text{Cyclopentane}$ | Q) Na/Ether |
| III) $\text{Acetone} \longrightarrow \text{Propan-2-ol}$ | R) $\text{NH}_2-\text{NH}_2 / \text{HO}^-$ |
| IV) $\text{CH}_3\text{Br} + \text{CH}_3\text{Br} \longrightarrow \text{C}_2\text{H}_6$ | S) $\text{Zn}-\text{Hg} / \text{HCl}$ |
- A) $I \rightarrow Q, II \rightarrow R, III \rightarrow P, IV \rightarrow Q$
 B) $I \rightarrow S, II \rightarrow R, III \rightarrow P, IV \rightarrow Q$
 C) $I \rightarrow Q, II \rightarrow P, III \rightarrow R, IV \rightarrow Q$
 D) $I \rightarrow S, II \rightarrow R, III \rightarrow P, IV \rightarrow Q$

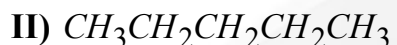




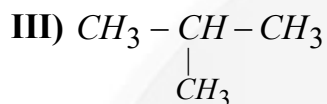
53. List – I contains alkane and List –II contains number of monochloro derivatives obtained (Including stereoisomers)



P) 2



Q) 3



R) 4

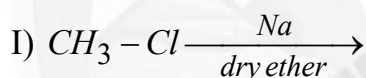


S) 5

A) $I \rightarrow P, II \rightarrow R, III \rightarrow P, IV \rightarrow Q$ B) $I \rightarrow P, II \rightarrow S, III \rightarrow P, IV \rightarrow Q$

C) $I \rightarrow P, II \rightarrow R, III \rightarrow P, IV \rightarrow P$ D) $I \rightarrow P, II \rightarrow S, III \rightarrow Q, IV \rightarrow Q$

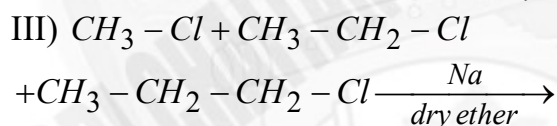
54. List – I contains Wurtz reaction & List - II contains number of coupled products (excluding stereo)



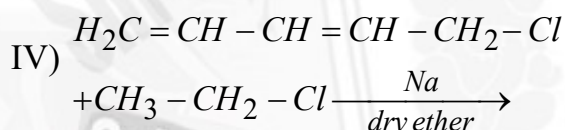
p) 5



q) 6



r) 3



s) 1

A) $I \rightarrow s, II \rightarrow r, III \rightarrow p, IV \rightarrow q$ B) $I \rightarrow s, II \rightarrow p, III \rightarrow r, IV \rightarrow q$

C) $I \rightarrow s, II \rightarrow q, III \rightarrow p, IV \rightarrow q$ D) $I \rightarrow s, II \rightarrow r, III \rightarrow q, IV \rightarrow p$





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10
RANKS (50%)

BELOW
100
All India Open
Category Ranks

32

BELOW
1000
All India Open
Category Ranks

181

BELOW
100
All India
Category Ranks
Count

89

BELOW
1000
All India
Category Ranks
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